Supporting Documentation: College Graduate Salary Analysis

Internal notes for fellow analysts: the model, its validation, and the code behind the article

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In the Glassdoor article, one linear regression model backs every section; the article's `.qmd` fits the model and pulls all provided numbers directly from it. This documentation follows this structure: **Section 1:** the model and its validation. **Section 2:** how each article section uses it. **Section 3:** the code.

1 Modeling

1.1 The model

773 graduates, 16 major categories, no missing values. We fit the model $\mathbf{y} \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2\mathbf{I})$ with $\mathbf{y}$ the 773×1 vector of salaries and $\mathbf{X}$ the 773×48 matrix shown in Table 1. For graduate i in major m:

$$\text{Salary}_i = \beta_0 + \alpha_m + (\beta_{\text{GPA}} + \gamma_m)\,\text{GPA}_i + (\beta_{\text{M}} + \delta_m)\,I(\text{Sex}_i = \text{M}) + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

Table 1: The 48 columns of the design matrix $\mathbf{X}$ ($\alpha_m = \gamma_m = \delta_m = 0$ for the baseline major).

Columns of $\mathbf{X}$	Count	Contents	Coefficient
Intercept	1	a column of ones	β_0
Major indicators	15	$I(\text{Major}_i = m)$; baseline is Agriculture & Natural Resources	α_m
GPA	1	the graduate's GPA (quantitative, not centered)	β_{GPA}
Male indicator	1	$I(\text{Sex}_i = \text{M})$; baseline is F	β_{M}
Major × GPA	15	$I(\text{Major}_i = m) \times \text{GPA}_i$	γ_m
Major × Male	15	$I(\text{Major}_i = m) \times I(\text{Sex}_i = \text{M})$	δ_m

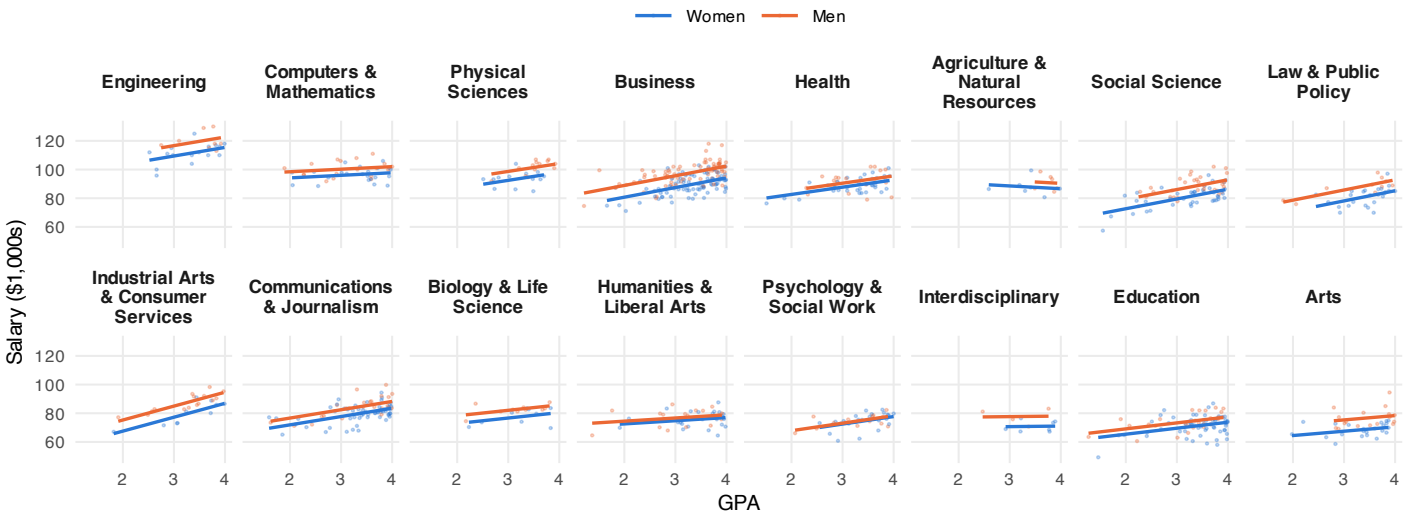


Figure 1: The fitted model. Each major has its own intercept, its own GPA slope, and its own gap between the lines for men and women. The two lines in a panel are parallel because there is no GPA by Sex term.

The two quantities the article leans on: the **GPA slope** in major m, $\beta_{\text{GPA}} + \gamma_m$, and the **men-minus-women gap** in major m, $\beta_{\text{M}} + \delta_m$.

1.2 Choosing the terms: nested F-tests

Table 2: Nested F-tests against the final model. A to E chose the terms; F to H test each variable overall.

Test	df1	df2	F	p	Decision
A. Same GPA slope in every major (drop Major:GPA)	15	725	2.01	0.013	Keep Major:GPA
B. Same sex gap in every major (drop Major:Sex)	15	725	2.13	0.0075	Keep Major:Sex
C. GPA slope differs by sex (add GPA:Sex)	1	724	0.16	0.69	Leave out
D. Full three-way model (add GPA:Sex and Major:GPA:Sex)	16	709	0.57	0.91	Leave out
E. Curved GPA effect (add GPA squared)	1	724	2.63	0.11	Leave out
F. No GPA effect (drop GPA and Major:GPA)	16	725	15.91	6.2e-38	GPA matters
G. No major effect (drop all 45 major terms)	45	725	55.14	3e-202	Major matters
H. No sex effect (drop Sex and Major:Sex)	16	725	13.22	2.8e-31	Sex matters

A and B are significant, but modest next to the main effects (F to H). The article is worded that way.

1.3 Checking the assumptions (LINE)

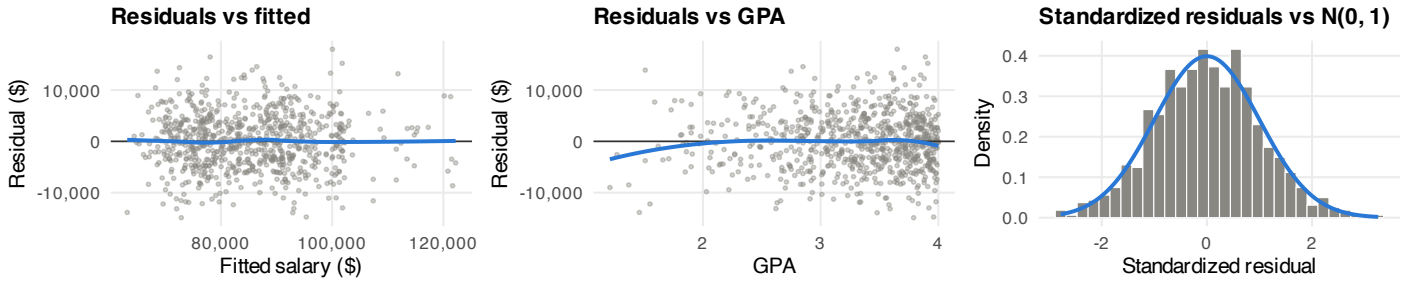


Figure 2: Residual diagnostics for the final model.

Assumption	Evidence	Verdict
Linearity	Residuals vs GPA flat above GPA 2 (middle); GPA^2 test E, $p = 0.11$	Holds (thin below GPA 2: 29 graduates)
Independence	One ACS respondent per row; no household or employer identifiers	Assumed by design; not testable here
Normality	Histogram follows N(0, 1) (right); KS test $p = 0.97$	Holds
Equal variance	No fanning (left); Breusch-Pagan $p = 0.71$	Holds; results stay in dollars
Influence	Largest Cook's distance 0.027; high leverage only where $n = 11, 13$	No concern; explains wide intervals in tiny majors

1.4 Validating predictions: 10-fold cross-validation

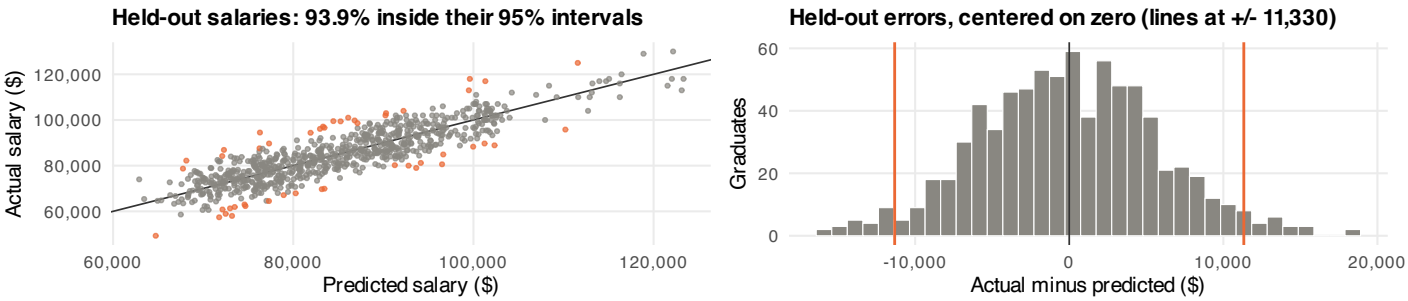


Figure 3: Final model, predicting graduates it had not seen (10-fold cross-validation). Orange points fell outside their 95% prediction interval; orange lines mark the average interval half-width.

Table 3: 10-fold cross-validated prediction performance (same folds for every model).

Model	RMSE	Bias	Coverage	Width	R2
Additive: GPA + Major + Sex	5,752	-1	0.942	22,547	0.774
Final: Major x GPA + Major x Sex	5,766	41	0.939	22,659	0.773
Three-way: Major x GPA x Sex	5,940	27	0.939	23,189	0.759

- **The prediction intervals are safe to quote.** Out-of-sample RMSE \$5,766 vs in-sample s = \$5,553; bias near zero; 95% intervals cover 93.9% of held-out salaries.
- **The interactions do not improve prediction.** The additive model predicts just as well. They stay because the article must say whether effects differ by major, and tests A and B say they do.

2 How the model backs each section of the article

Every estimate is a linear combination $\mathbf{a}'\hat{\beta} \pm t_{0.975,\,n-p}\, s\sqrt{\mathbf{a}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{a}}$ from `multcomp::glht()`. Each $\mathbf{a}$ is built from rows of $\mathbf{X}$ for hypothetical graduates; the sex gap, for example, is (row for a man) minus (row for a woman) with the same major and GPA.

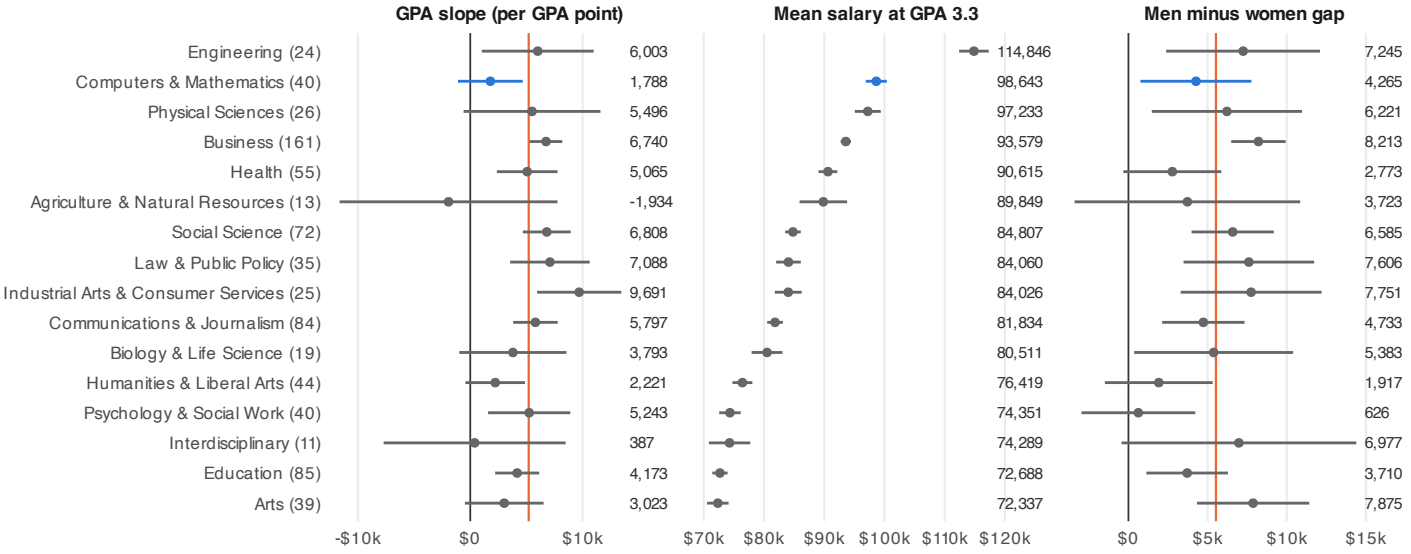


Figure 4: Estimates by major with 95% confidence intervals, in dollars (number of graduates in parentheses; our major in blue). In the slope and gap panels, the dark line is zero and the orange line is the sample-weighted average over majors.

Table 4: Where each claim in the article comes from.

The article says	Model quantity or hypothesis	How	Result
Section 2: GPA			
GPA matters	H_0: $\beta_{\text{GPA}} = 0$ and every $\gamma_m = 0$	F-test F	F = 15.91, p = 6.2e-38
"About \$5,200 per GPA point"	$\sum_m w_m(\beta_{\text{GPA}} + \gamma_m)$, with w_m = major's share of graduates	Contrast, 95% CI	\$5,202 (\$4,473, \$5,931)
The payoff differs by major	H_0: every $\gamma_m = 0$; slopes $\beta_{\text{GPA}} + \gamma_m$	F-test A; Figure 4, left	F = 2.01, p = 0.013
No need to be hyper-anxious	Judgment: slope vs the spread of salaries and vs the gap between majors	s; R^2 of GPA alone	s = \$5,553; R^2 = 0.052
Section 3: Majors			
Majors differ	H_0: all 45 major terms are 0	F-test G	F = 55.14, p = 3e-202
The ranking of majors	Mean at GPA 3.3, women and men averaged: $\frac{1}{2}[\mathbf{x}'(m,\text{F}) + \mathbf{x}'(m,\text{M})]\beta$	Contrasts, 95% CIs; Figure 4, middle	Engineering \$114,846 down to Arts \$72,337
Top to bottom is "about \$42,500"	Engineering minus Arts	Contrast, 95% CI	\$42,509 (\$39,468, \$45,550), or 8.2 average GPA points
Section 4: Sex gap			

The article says	Model quantity or hypothesis	How	Result
A gap exists within majors	H_0: $\beta_{\mathrm{M}} = 0$ and every $\delta_m = 0$	F-test H	F = 13.22, p = 2.8e-31
The gap differs by major	H_0: every $\delta_m = 0$; gaps $\beta_{\mathrm{M}} + \delta_m$	F-test B; Figure 4, right	F = 2.13, p = 0.0075
"About $5,500 a year" on average	$\sum_m w_m(\beta_{\mathrm{M}} + \delta_m)$	Contrast, 95% CI	$5,527 ($4,704, $6,349)
Our major as the example	Computers & Mathematics: means for women and men at GPA 3.3, and their difference	Contrasts, 95% CI	Women $96,510, men $100,776; gap $4,265 ($760, $7,770)
Section 5: Other factors			
"About 80%" of salary differences explained	R^2 (adjusted; cross-validated)	`summary(fit)`; Section 1.4	0.802 (0.789; 0.773)
Major does most of the work	R^2 of each variable alone	Three one-variable fits	Major 0.674, sex 0.068, GPA 0.052
The salary range for one person	95% *prediction* interval: a woman in our major with a 3.5 GPA	`predict(interval = "prediction")`	$96,868 ($85,707, $108,029)
"Typically off by about $5,800"; ranges right "94% of the time"	Cross-validated RMSE and interval coverage	Section 1.4	$5,766; 93.9%

Caveats we wrote into the article

- **Association, not causation.** The slopes compare different graduates; the article says "goes with", not "causes".
- **Small majors are fuzzy.** Several slope intervals include zero, so the article warns against reading much into any one small major. Majors are described in tiers; we did not run all 120 pairwise comparisons.
- **Multiplicity.** The gap intervals are one-at-a-time: 11 of 16 exclude zero, but only 6 do with `multcomp`'s simultaneous adjustment, and our own major's then crosses zero. The article gives its estimate and range without calling it proven.
- **The gap is not "unequal pay for identical work."** It conditions on major and GPA only, not on job, industry, hours, or location. It is the same at every GPA (test C, $p = 0.69$). The list of other factors in Section 5 is subject-matter reasoning.

3 Code

All of the code behind this document, in the order it runs. The article's own figures are drawn in `salary_blog.qmd` from this same model and these same estimates.

```
# ==== Section 1.1: data, the model, and helpers ====
library(dplyr); library(ggplot2); library(patchwork)

# Data: one row per graduate. Factor levels are alphabetical, so the baselines in X
# are "Agriculture & Natural Resources" and "F".
dat <- vroom::vroom("Salary.csv", show_col_types = FALSE) |>
  mutate(MajorCategory = factor(MajorCategory), Sex = factor(Sex, levels = c("F", "M")))

majors    <- levels(dat$MajorCategory)
n_major   <- as.numeric(table(dat$MajorCategory))
w         <- n_major / sum(n_major)       # each major's share of graduates
our_major <- "Computers & Mathematics"    # statistics falls in this category
gpa_ref   <- 3.3                          # typical GPA used for comparisons (mean = 3.30)

# The one model behind the whole article
f_final <- Salary ~ MajorCategory * GPA + MajorCategory * Sex
fit     <- lm(f_final, data = dat)
X       <- model.matrix(fit)              # the 773 x 48 design matrix of Section 1.1

# Formatting helpers: numbers with commas, dollars for the text, "estimate (lower, upper)"
# strings, and one table style (LaTeX booktabs, held in place, small font)
comma  <- function(x) format(round(x), big.mark = ",", trim = TRUE)
usd    <- function(x) paste0("\\$", comma(x))
ci_usd <- function(d) sprintf("%s (%s, %s)", usd(d$Estimate), usd(d$Lower), usd(d$Upper))
tbl <- function(x, caption, ...) {
  caption <- gsub("%", "\\%", caption, fixed = TRUE)   # kable() does not escape captions
  out <- knitr::kable(x, format = "latex", booktabs = TRUE, position = "H", linesep = "",
                      label = NA, caption = caption, ...)
  knitr::asis_output(sub("\\centering", "\\centering\\small", out, fixed = TRUE))
```

```
}

# One look for every figure: blue = accent, orange = second series, gray = context
blue <- "#2a78d6"; orange <- "#eb6834"; gray <- "#898781"
theme_doc <- function() {
  theme_minimal(base_size = 8) +
    theme(panel.grid.minor = element_blank(), legend.position = "top",
          legend.margin = margin(0, 0, 0, 0), legend.box.spacing = unit(2, "pt"),
          plot.title = element_text(size = 8.5, face = "bold"),
          strip.text = element_text(size = 7, face = "bold"))
}
```

```
# ==== Figure 1: the fitted model, one panel per major ====
# Majors ranked by the model's mean salary at GPA 3.3 (women and men averaged);
# the same order is used again in Figure 4
ref_mean    <- function(sex) predict(fit, data.frame(MajorCategory = majors, Sex = sex, GPA = gpa_ref))
major_order <- majors[order(-(ref_mean("F") + ref_mean("M")) / 2)]

dat |>
  mutate(Fitted = fitted(fit), Major = factor(MajorCategory, levels = major_order),
         Sex = factor(ifelse(Sex == "F", "Women", "Men"), levels = c("Women", "Men"))) |>
  ggplot(aes(GPA, Salary / 1000, color = Sex)) +
  geom_point(alpha = 0.3, size = 0.3) +                      # the graduates
  geom_line(aes(y = Fitted / 1000), linewidth = 0.6) +       # the fitted line for each sex
  facet_wrap(~ Major, nrow = 2, labeller = label_wrap_gen(width = 16)) +
  scale_color_manual(values = c(Women = blue, Men = orange)) +
  labs(x = "GPA", y = "Salary ($1,000s)", color = NULL) +
  theme_doc()
```

```
# ==== Section 1.2: nested F-tests ====
# One nested F-test (reduced vs full model) as a table row
nested_f <- function(test, reduced, full = f_final) {
  a <- anova(lm(reduced, data = dat), lm(full, data = dat))
  data.frame(Test = test, df1 = a$Df[2], df2 = a$Res.Df[2], F = a$F[2], p = a$`Pr(>F)`[2])
}
ftests <- rbind(
  # A-E: which terms belong in the model?
  nested_f("A. Same GPA slope in every major (drop Major:GPA)", Salary ~ GPA + MajorCategory * Sex),
  nested_f("B. Same sex gap in every major (drop Major:Sex)", Salary ~ MajorCategory * GPA + Sex),
  nested_f("C. GPA slope differs by sex (add GPA:Sex)", f_final, update(f_final, . ~ . + GPA:Sex)),
  nested_f("D. Full three-way model (add GPA:Sex and Major:GPA:Sex)", f_final,
           Salary ~ MajorCategory * GPA * Sex),
  nested_f("E. Curved GPA effect (add GPA squared)", f_final, update(f_final, . ~ . + I(GPA^2))),
  # F-H: does each variable matter at all? (used in Section 2)
  nested_f("F. No GPA effect (drop GPA and Major:GPA)", Salary ~ MajorCategory * Sex),
  nested_f("G. No major effect (drop all 45 major terms)", Salary ~ GPA + Sex),
  nested_f("H. No sex effect (drop Sex and Major:Sex)", Salary ~ MajorCategory * GPA)
)
ftests$Decision <- c("Keep Major:GPA", "Keep Major:Sex", "Leave out", "Leave out", "Leave out",
                     "GPA matters", "Major matters", "Sex matters")

# F statistics and p-values by letter, for the text and for Table 4
Fv    <- setNames(ftests$F, LETTERS[1:8])
pv    <- setNames(ftests$p, LETTERS[1:8])
fmt_p <- function(p) sapply(p, function(v) format(signif(v, 2)))
ftxt  <- function(l) sprintf("F = %.2f, p = %s", Fv[[l]], fmt_p(pv[[l]]))

ftests |>
  mutate(F = sprintf("%.2f", F), p = fmt_p(p)) |>
  tbl("Nested F-tests against the final model. A to E chose the terms; F to H test each variable overall.",
      align = "lrrrrl")
```

```
# ==== Section 1.3: assumption checks ====
diag_df <- data.frame(fitted = fitted(fit), resid = resid(fit), GPA = dat$GPA,
                      std_resid = MASS::stdres(fit))      # standardized residuals
bp_test <- lmtest::bptest(fit)                            # Breusch-Pagan, H0: equal variance
ks_test <- ks.test(diag_df$std_resid, "pnorm")            # Kolmogorov-Smirnov, H0: normal

# Residuals against x with a loess curve: a flat curve on zero means a straight line is enough
resid_plot <- function(x, xlab, title) {
  ggplot(diag_df, aes({{ x }}, resid)) +
    geom_point(color = gray, alpha = 0.4, size = 0.5) +
    geom_hline(yintercept = 0, color = "grey20", linewidth = 0.3) +
    geom_smooth(se = FALSE, method = "loess", color = blue, linewidth = 0.7) +
```

```
    scale_x_continuous(labels = scales::label_comma()) +
    scale_y_continuous(labels = scales::label_comma()) +
    labs(title = title, x = xlab, y = "Residual ($)") +
    theme_doc()
}
# Histogram of standardized residuals with the N(0, 1) density on top
p_hist <- ggplot(diag_df, aes(std_resid)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = gray, color = "white",
                 linewidth = 0.2) +
  stat_function(fun = dnorm, color = blue, linewidth = 0.7) +
  labs(title = "Standardized residuals vs N(0, 1)", x = "Standardized residual", y = "Density") +
  theme_doc()

resid_plot(fitted, "Fitted salary ($)", "Residuals vs fitted") +
  resid_plot(GPA, "GPA", "Residuals vs GPA") + p_hist
```

```
# ==== Section 1.4: 10-fold cross-validation of the predictions ====
set.seed(469)
k     <- 10
folds <- sample(cut(seq(1, nrow(dat)), breaks = k, labels = FALSE))   # random fold labels
# Each training set must keep every Major x Sex cell, or it could not estimate all 48 betas
stopifnot(sapply(1:k, function(i) all(table(dat$MajorCategory[folds != i], dat$Sex[folds != i]) > 0)))

# Out-of-sample predictions with 95% prediction intervals: fit on k - 1 folds, predict the other
cv_predict <- function(form) {
  out <- lapply(1:k, function(i) {
    f <- lm(form, data = dat[folds != i, ])
    cbind(Salary = dat$Salary[folds == i],
          predict(f, newdata = dat[folds == i, ], interval = "prediction", level = 0.95))
  })
  as.data.frame(do.call(rbind, out))
}
# RMSE = typical miss, Bias = systematic over/under, Coverage = share of held-out salaries
# inside the 95% interval (want 0.95), Width = average interval width, R2 = out-of-sample R^2
cv_summary <- function(model, cv) {
  data.frame(Model = model, RMSE = sqrt(mean((cv$Salary - cv$fit)^2)), Bias = mean(cv$fit - cv$Salary),
             Coverage = mean(cv$Salary >= cv$lwr & cv$Salary <= cv$upr), Width = mean(cv$upr - cv$lwr),
             R2 = 1 - sum((cv$Salary - cv$fit)^2) / sum((cv$Salary - mean(cv$Salary))^2))
}
cv_pred  <- cv_predict(f_final)            # held-out predictions from the final model
cv_table <- rbind(
  cv_summary("Additive: GPA + Major + Sex", cv_predict(Salary ~ GPA + MajorCategory + Sex)),
  cv_summary("Final: Major x GPA + Major x Sex", cv_pred),
  cv_summary("Three-way: Major x GPA x Sex", cv_predict(Salary ~ MajorCategory * GPA * Sex)))
cv_final <- cv_table[2, ]

# Left: held-out salary vs prediction, flagged by whether it fell inside its 95% interval.
# Right: the held-out errors, with the average interval half-width marked.
cv_pred <- cv_pred |>
  mutate(error = Salary - fit, Interval = ifelse(Salary >= lwr & Salary <= upr, "Inside", "Outside"))
half <- mean(cv_pred$upr - cv_pred$fit)
p_cv <- ggplot(cv_pred, aes(fit, Salary, color = Interval)) +
  geom_abline(slope = 1, intercept = 0, color = "grey20", linewidth = 0.3) +
  geom_point(alpha = 0.7, size = 0.6) +
  scale_color_manual(values = c(Inside = gray, Outside = orange), guide = "none") +
  scale_x_continuous(labels = scales::label_comma()) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(title = sprintf("Held-out salaries: %.1f%% inside their 95%% intervals", 100 * cv_final$Coverage),
       x = "Predicted salary ($)", y = "Actual salary ($)") +
  theme_doc()
p_err <- ggplot(cv_pred, aes(error)) +
  geom_histogram(bins = 35, fill = gray, color = "white", linewidth = 0.2) +
  geom_vline(xintercept = c(-half, half), color = orange, linewidth = 0.5) +
  geom_vline(xintercept = 0, color = "grey20", linewidth = 0.3) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(title = paste0("Held-out errors, centered on zero (lines at +/- ", comma(half), ")"),
       x = "Actual minus predicted ($)", y = "Graduates") +
  theme_doc()
p_cv + p_err

# Table 3: the same summaries for the final model and two competitors
tbl(cv_table, "10-fold cross-validated prediction performance (same folds for every model).",
    digits = c(0, 0, 0, 3, 0, 3), format.args = list(big.mark = ","))
```

```r
# ==== Section 2: contrasts, confidence intervals, and the prediction example ====
# Row(s) of X for hypothetical graduates; each contrast below is a difference or average of rows
xrow <- function(major, sex, gpa) {
  newdata <- data.frame(MajorCategory = factor(major, levels = majors),
                        Sex = factor(sex, levels = c("F", "M")), GPA = gpa)
  model.matrix(delete.response(terms(fit)), newdata)
}
# Estimate and ordinary (one-at-a-time) 95% t interval for each row a' of K
lin_combo <- function(K, label) {
  ci <- confint(multcomp::glht(fit, linfct = K), calpha = multcomp::univariate_calpha())$confint
  data.frame(Label = label, Estimate = ci[, "Estimate"], Lower = ci[, "lwr"], Upper = ci[, "upr"],
             row.names = NULL)
}
K_slope <- xrow(majors, "F", 1) - xrow(majors, "F", 0)                    # beta_GPA + gamma_m
K_gap   <- xrow(majors, "M", gpa_ref) - xrow(majors, "F", gpa_ref)        # beta_M + delta_m
K_mean  <- (xrow(majors, "F", gpa_ref) + xrow(majors, "M", gpa_ref)) / 2  # mean salary at GPA 3.3

slopes      <- lin_combo(K_slope, majors) |> mutate(n = n_major)
gaps        <- lin_combo(K_gap, majors)   |> mutate(n = n_major)
major_means <- lin_combo(K_mean, majors)  |> mutate(n = n_major)
avg_slope   <- lin_combo(t(w) %*% K_slope, "Average slope")   # sample-weighted over majors
avg_gap     <- lin_combo(t(w) %*% K_gap, "Average gap")       # sample-weighted over majors
top <- which.max(major_means$Estimate); bottom <- which.min(major_means$Estimate)
top_vs_bottom <- lin_combo(K_mean[top, , drop = FALSE] - K_mean[bottom, , drop = FALSE], "Top - bottom")

# Our major: mean salary for women and for men at GPA 3.3, and the gap between them
ours_f   <- lin_combo(xrow(our_major, "F", gpa_ref), "Women")
ours_m   <- lin_combo(xrow(our_major, "M", gpa_ref), "Men")
ours_gap <- gaps[majors == our_major, ]

# The same 16 gaps with simultaneous (family-wise 95%) intervals, to gauge multiplicity
set.seed(469)                               # the adjusted critical value is simulated
adj_lwr <- confint(multcomp::glht(fit, linfct = K_gap))$confint[, "lwr"]

# Prediction example (article Section 5) and R-squared of the pieces
example <- data.frame(MajorCategory = our_major, Sex = "F", GPA = 3.5)
ex_pi   <- predict(fit, example, interval = "prediction", level = 0.95)   # one new graduate
r2      <- function(form) summary(lm(form, data = dat))$r.squared

# Figure 4: the three sets of estimates side by side, majors ranked by mean salary
forest <- bind_rows("GPA slope (per GPA point)" = slopes, "Mean salary at GPA 3.3" = major_means,
                    "Men minus women gap" = gaps, .id = "panel") |>
  mutate(panel = factor(panel, levels = unique(panel)),
         Major = reorder(paste0(Label, " (", n, ")"), -match(Label, major_order)),
         ours  = ifelse(Label == our_major, "ours", "other")) |>
  group_by(panel) |>
  mutate(label_x = max(Upper) + 0.03 * (max(Upper) - min(Lower))) |>   # where the numbers go
  ungroup()
refs <- data.frame(panel = factor(levels(forest$panel)[c(1, 3)], levels = levels(forest$panel)),
                   zero = 0, average = c(avg_slope$Estimate, avg_gap$Estimate))

ggplot(forest, aes(y = Major, color = ours)) +
  geom_vline(data = refs, aes(xintercept = zero), color = "grey20", linewidth = 0.3,
             inherit.aes = FALSE) +
  geom_vline(data = refs, aes(xintercept = average), color = orange, linewidth = 0.4,
             inherit.aes = FALSE) +
  geom_linerange(aes(xmin = Lower, xmax = Upper), linewidth = 0.5) +
  geom_point(aes(x = Estimate), size = 1.2) +
  geom_text(aes(x = label_x, label = comma(Estimate)), hjust = 0, size = 2.1, color = "grey20") +
  facet_wrap(~ panel, nrow = 1, scales = "free_x") +
  scale_color_manual(values = c(ours = blue, other = "grey40"), guide = "none") +
  scale_x_continuous(labels = \(x) ifelse(x == 0, "$0", scales::dollar(x / 1000, suffix = "k")),
                     expand = expansion(mult = c(0.03, 0.2))) +
  labs(x = NULL, y = NULL) +
  theme_doc() + theme(panel.grid.major.y = element_blank())
```